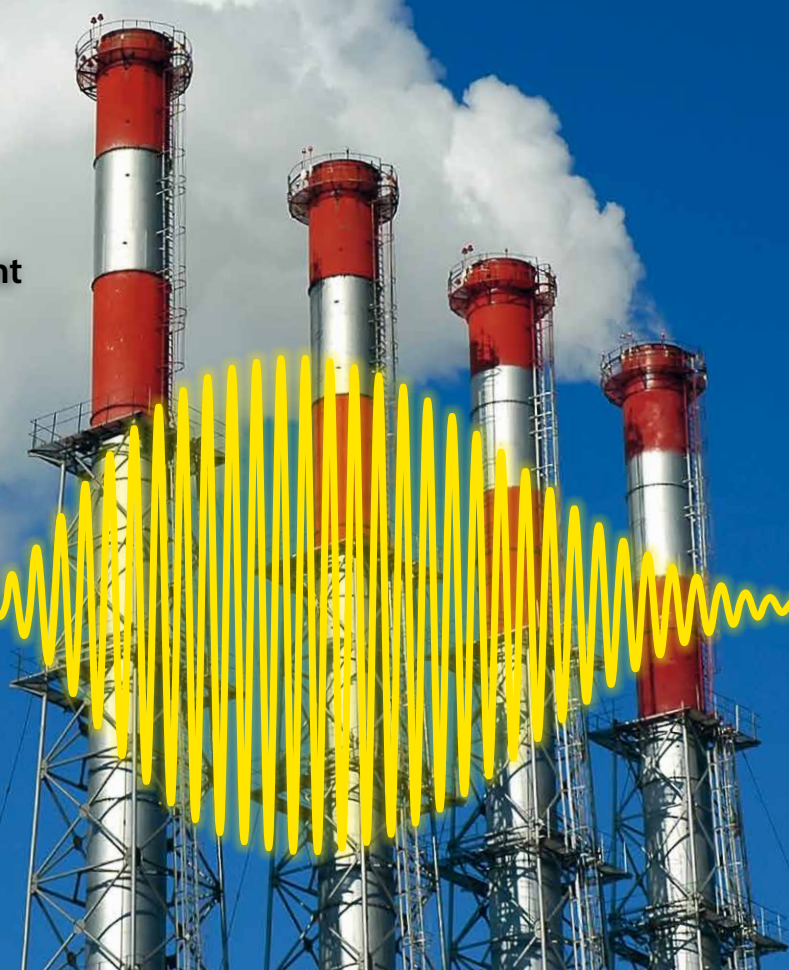


D-FL 220

Ultrasonic Volume Flow Measuring System

- In situ measurement, continuous measurement
- Measurement even in wet or dust-laden flue gases
- Allows calibration to measure the volume flow
- Automatic zero and reference point check
- Easy installation, adjustment and maintenance
- Very low maintenance
- Remote access possible



Volume Flow Measuring System

The new D-FL 220 allows for reliable measurement of the velocity and volume

Ultrasonic sensor system to measure the velocity and the volume flow of flue gas or process gas in pipes or ducts.

Features

- In situ measurement, continuous measurement
- Contactless measurement without any moving parts
- Operation via USB or Modbus RTU interface, optional with the universal control unit D-ISC 100
- Data transfer via Modbus RTU interface according to VDI 4201-3

- Remote access possible
- Measurement even in wet or dust-laden flue gas
- Allows calibration to measure the volume flow
- Automatic zero and reference point check
- Easy installation, adjustment and maintenance
- Very low maintenance

Applications

- Volume flow measurements at low speed
- Plants with damp and/or aggressive exhaust gas, e.g. in waste incineration plants
- Volume flow measurement at high dust content
- Incinerators in general
- Power plants, heating plants, cement plants
- Chemical and petrochemical industries
- Pulp and paper industry
- Biomass incinerators



Refinery



Cracker



Chemical Industry

flow of flue gas or process gas in a broad range of applications.

Operating principle

The flow velocity of sample gas is precisely calculated from the transit time difference dependant on the direction. Taking into account the cross-section of the stack, the gas temperature and absolute pressure the flow rate is calculated.

D-FL 220 performs automatic check cycles and is extremely low maintenance.

System components

System components:

- Measuring heads
- Purge flanges (material 1.4571)
- Welding pipe with flange (1.4571, FRP)
- Purge air unit for cleaning and cooling the sensors
- Terminal box for power supply of the sensors
- Universal control unit D-ISC 100 optional

The D-FL 220 can either be operated through local USB port or remotely using the RS 485 bidirectional Modbus RTU interface. Optional the universal control unit D-ISC 100 can be used.

Universal control unit D-ISC 100

The universal control unit allows the connection of up to eight measuring heads and allows a combination of dust and volume flow measurement. The D-ISC 100 offers remote inquiry and easy parameterization of connected DURAG devices. The display allows direct view of the current measuring values.

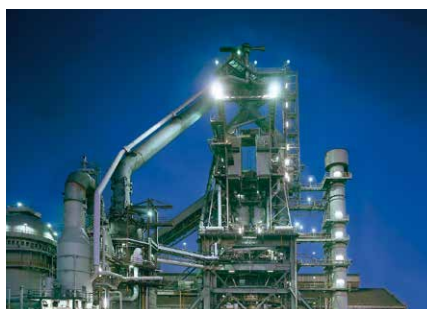
- Connection of up to 8 measuring heads
- Operation with keypad, through local USB port or remotely using RS 485 Modbus RTU interface
- Allows remote access to the connected measuring heads
- Expandable with extension modules



Technical data	
Measurements	flow velocity, volume flow, at operating conditions or normalized
Sample gas	air, flue gas, non flammable process gas
Flow velocity	0 .. 40 m/s
Duct diameter	0.5 – 13 m
Sample gas temperature	-20 to 120 / 150 / 200 / 250 / 300°C, others on request
Sample gas pressure (relative to ambient pressure)	-50 to +20 hPa, others on request
Sample gas moisture	0 – 100 % rel. humidity, condensation allowed
Ambient temperature	-20 to 50 °C, others on request



Steelworks



Blast furnace



Steam reformer